

How COVID-19 pandemic impacted global economy

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Outline

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Background

The COVID-19 outbreak began in late 2019 and quickly escalated into a global pandemic, prompting widespread lockdowns around the world. These restrictions forced many businesses, particularly non-essential businesses, to shut down, significantly affecting the global economy and the U.S. economy. The pandemic disrupted production, international trade, and employment across numerous countries, including the United States. This report highlights the pandemic's impact on key economic performance indicators, with a particular focus on inflation and unemployment rates. It begins with an overview of global inflation trends before narrowing its focus to the United States.

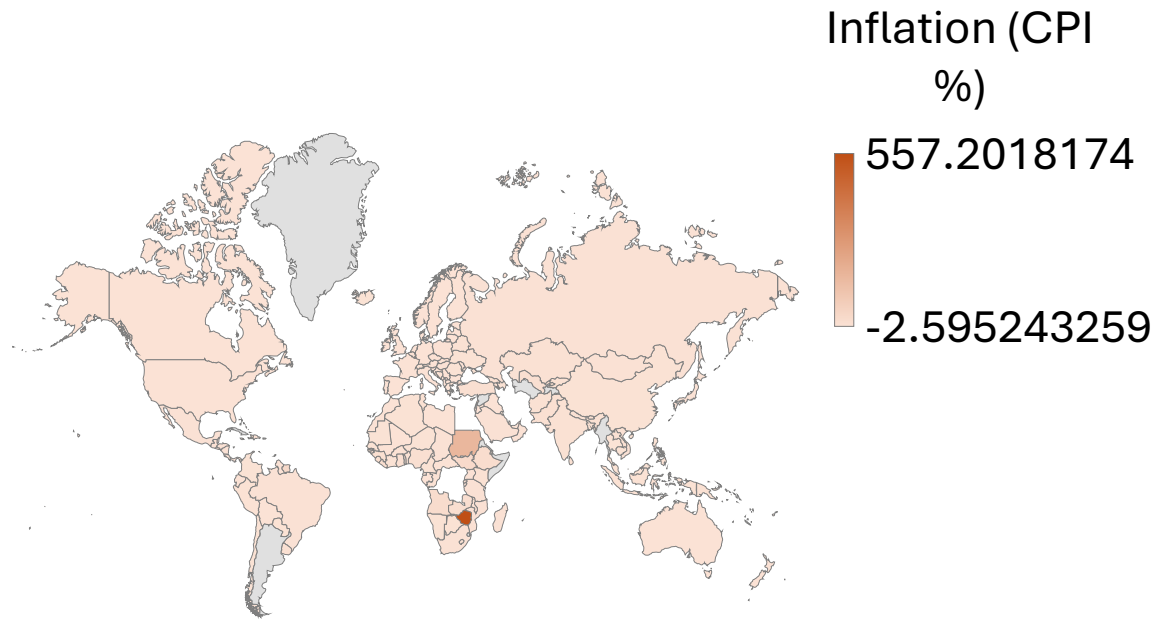
Data and Method

The data used in this report were obtained from Kaggle and include indicators such as inflation, GDP, GDP per capita, unemployment rate, interest rate, and gross national income. To assess the economic effects of the COVID-19 pandemic, the data were categorized into three periods: the pre-pandemic period (2017–2019), the pandemic period (2020–2022), and the post-pandemic period (2023–2025). This classification enables a comparison of economic performance before, during, and after the pandemic.

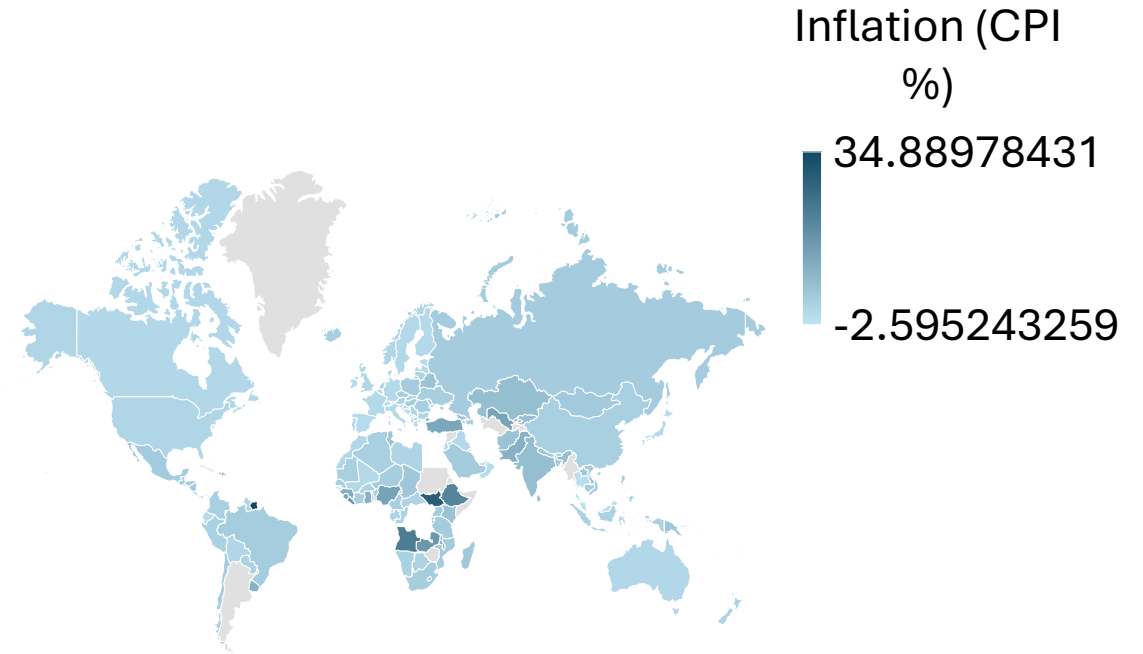
Key Findings

Global inflation rate during the pandemic (2020)

(A) Global inflation rate (CPI %) during the pandemic (2020)



(B) Global inflation rate without the outliers

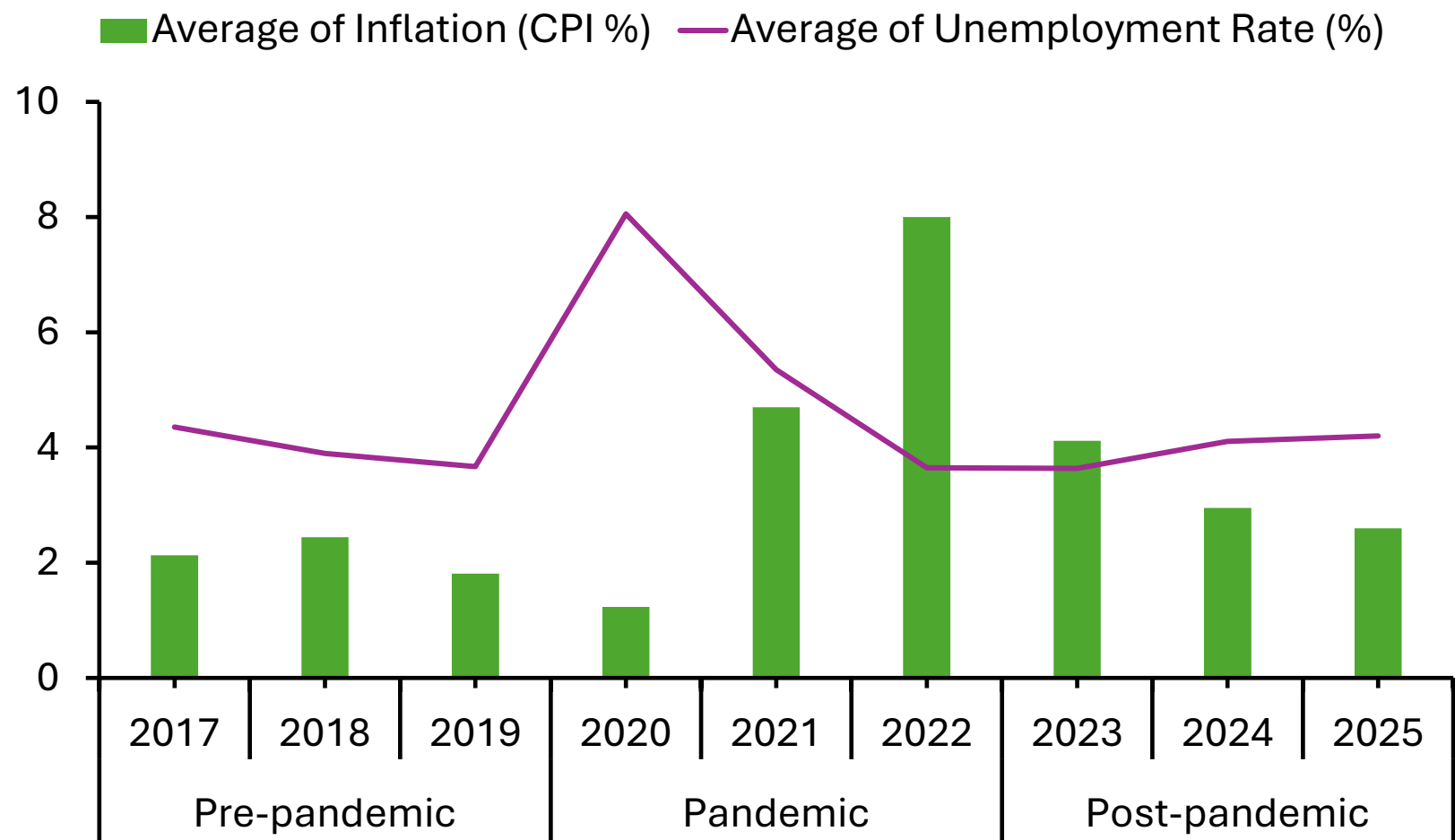


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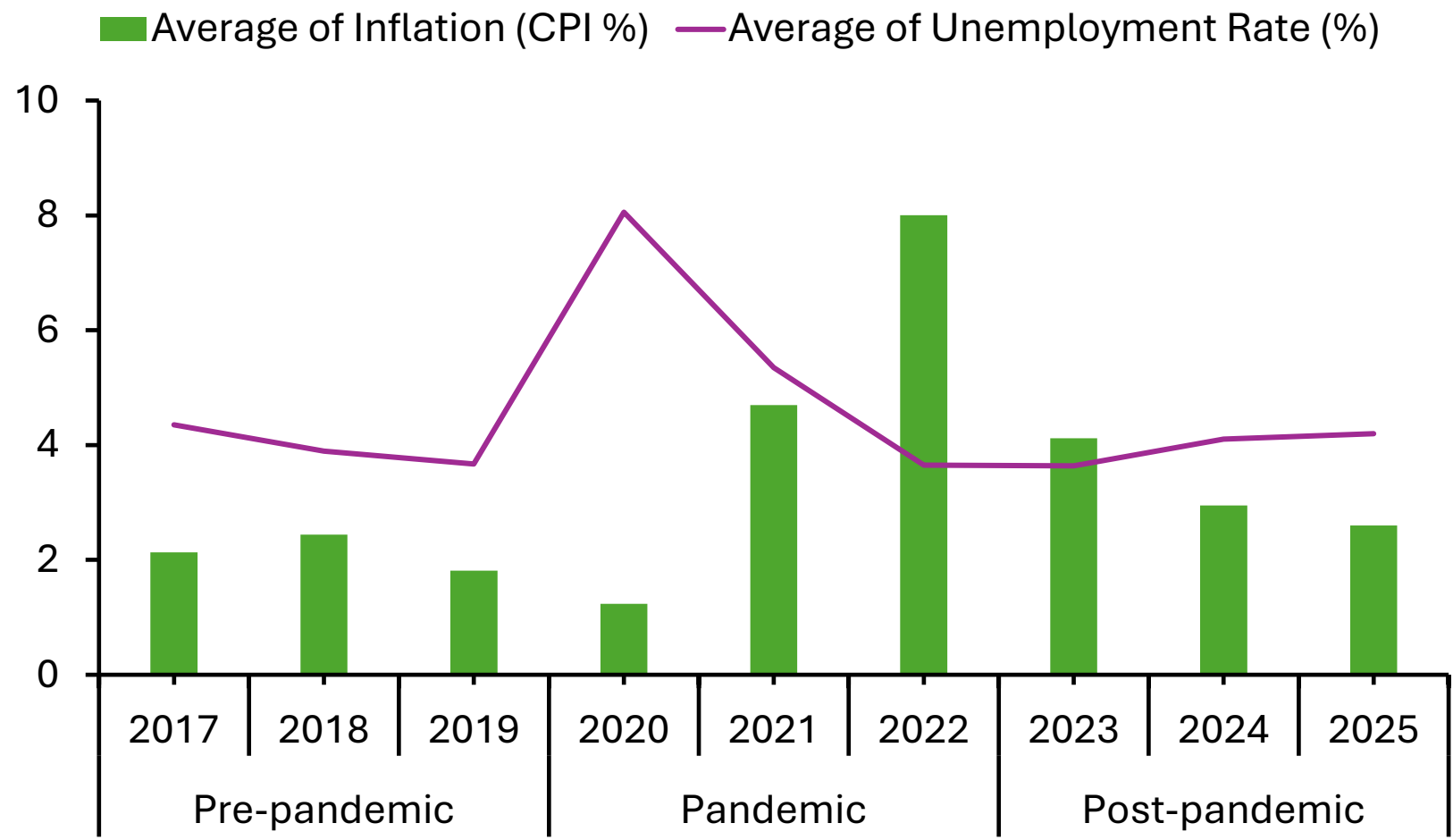
At the onset of the COVID-19 pandemic, the highest average inflation rates were observed in Lebanon (84%), Sudan (163%), and Zimbabwe (557%). After excluding these outliers, the highest inflation rates were concentrated in Africa and South America, likely reflecting pre-existing economic conditions. Data were missing for approximately 40 countries, including Argentina, the Democratic Republic of the Congo, and Venezuela; therefore, these findings do not represent all countries worldwide.

Inflation rate across three eras of COVID-19 pandemic in the US.



The average inflation rate increased by 119%, from 2.1% before the pandemic to 4.6% during the pandemic. It then declined to 3.2% in the post-pandemic period but remained 52% above the pre-pandemic level.

Unemployment rate across three eras of COVID-19 pandemic in the US.



The average unemployment rate increased by 43%, from 4.0% before the pandemic to 5.7% during the pandemic. It then declined to 4.0% in the post-pandemic period, returning to its pre-pandemic level.

Key performance indicators for the three eras of pandemic unemployment rate

Row Labels	Average of Unemployment Rate (%)	Average of Unemployment Rate (%) 2 Status
Pre-pandemic	3.973333333	●
Pandemic	5.684666667	●
Post-pandemic	3.981333333	●
Grand Total	4.546444444	●

This KPI highlights unemployment rates across the pre-pandemic, pandemic, and post-pandemic periods, as shown in the line graph on the previous slide.

Conclusion

- High inflation rates were concentrated in Africa and South America. However, missing data from approximately 40 countries limited a complete assessment of global inflation at the onset of the pandemic.
- The pandemic increased both inflation and unemployment in the United States. While unemployment has returned to its pre-pandemic level, inflation remains above its pre-pandemic rate.

Lessons learned

- During this project, I learned how to use Power Query to import, clean, and prepare data for analysis. I also used Pivot Table and Power Pivot to summarize data and create key performance indicators, skills that can be applied to future projects.
- In addition, I strengthened my Excel skills by using filtering, sorting, and other formulas. I also used data validation and the FILTER function to analyze the effects of the pandemic on inflation and unemployment.
- I also learned the importance of examining countries' economic conditions before the pandemic to better assess its impact on the key performance indicators presented in this report
- In future projects, I would use a more complete dataset to provide a more comprehensive analysis.